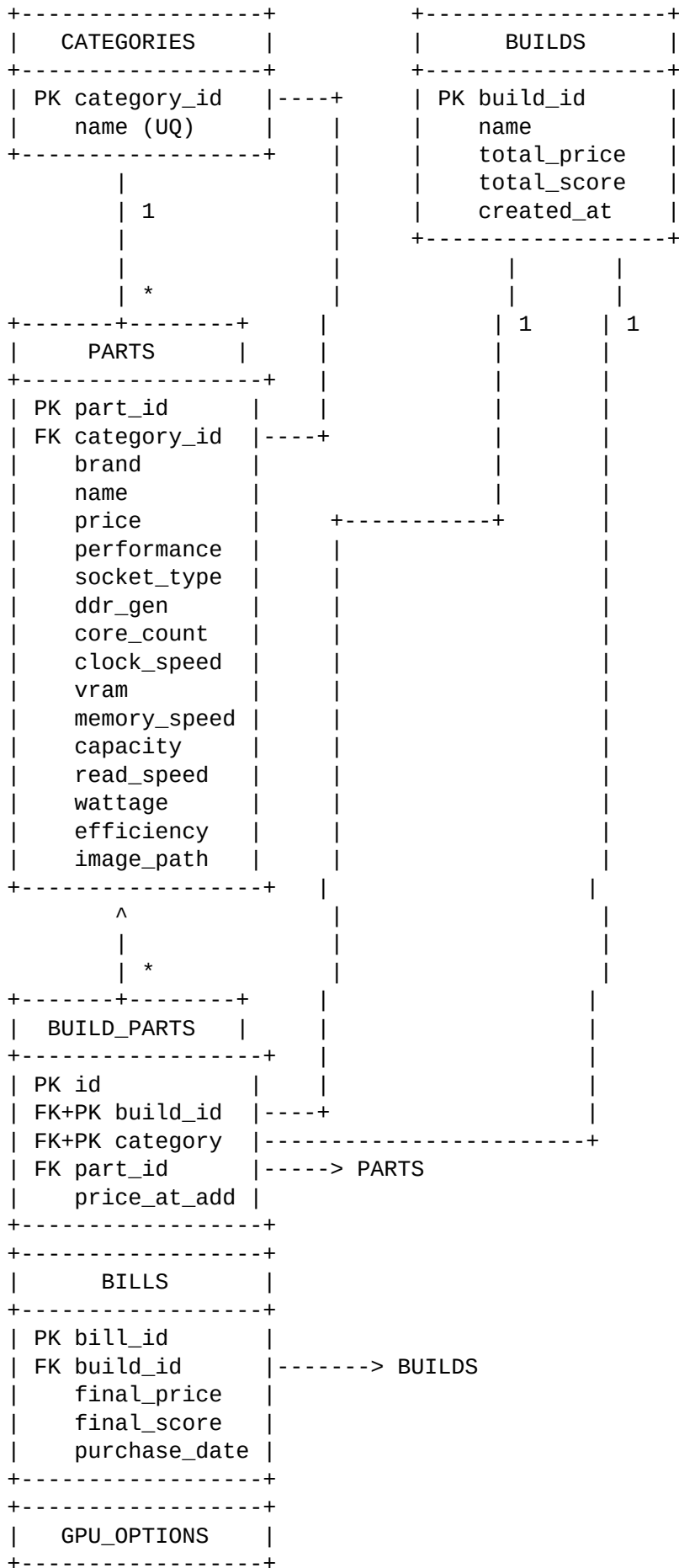


ER DIAGRAM - PC Build Store Database



```

| PK gpu_option_id |
| FK gpu_part_id   |-----> PARTS
|   for_budget     |
|   price_incr     |
|   perf_incr      |
+-----+

```

RELATIONSHIP SUMMARY

Table	FK Column	References	Cardinality
parts	category_id	categories	Many : 1
build_parts	build_id	builds	Many : 1
build_parts	category_id	categories	Many : 1
build_parts	part_id	parts	Many : 1
bills	build_id	builds	Many : 1
gpu_options	gpu_part_id	parts	Many : 1

UNIQUE CONSTRAINT on build_parts(build_id, category_id):
Each build can have at most ONE part per category.

ENTITY DESCRIPTIONS

CATEGORIES

Stores the five hardware component types. Each part belongs to exactly one category. The category_id is used throughout the system to distinguish between CPU, GPU, RAM, Storage, and PSU.

PARTS

The central entity containing all hardware parts. Uses a wide table design where most columns are nullable - only category-specific columns are populated per row. Columns map to category-specific attributes:

```

CPU    -> socket_type, ddr_generation, core_count, clock_speed
GPU    -> vram
RAM    -> ddr_generation, memory_speed, capacity
SSD    -> capacity, read_speed
PSU    -> wattage, efficiency

```

BUILDS

Represents a PC build configuration. The total_price and total_score are denormalized aggregates that are recalculated whenever build_parts change. Each build can have multiple parts assigned via build_parts.

BUILD_PARTS

Junction table implementing the many-to-many relationship between builds and parts, constrained so each build has at most one part per category. The price_at_add column captures a price snapshot so historical build costs remain accurate even if part prices change.

BILLS

Records a finalized purchase. Each bill references a single build and stores the final_price and final_score at the time of purchase. The purchase_date is auto-set to the current timestamp.

GPU_OPTIONS

Defines GPU upgrade paths for builds. Each option specifies a target GPU part, the budget tier it applies to, and the resulting price and performance deltas. Used by the GPUUpgradesGUI to suggest upgrades.

